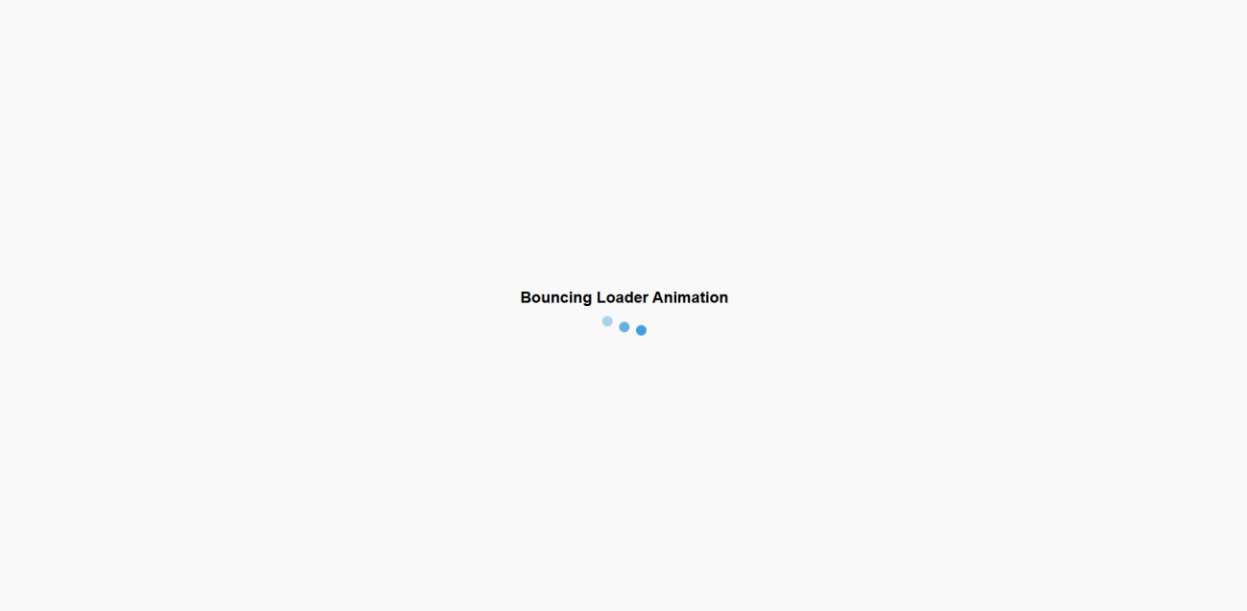
LAB-6

TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	—	Declares the document as an HTML5 document.
<html>	lang="en"	Root element of the HTML document and specifies the language as English.
<head>	—	Contains metadata and information about the HTML document.
<meta>	charset="UTF-8"	Specifies the character encoding of the document as UTF-8.
<meta>	name="viewport", content="width=device-width, initial-scale=1.0"	Makes the webpage responsive on different screen sizes.
<title>	—	Specifies the title shown in the browser tab.
<link>	rel="stylesheet", href="style1.css"	Links the external CSS file style1.css to the HTML document.
<body>	—	Contains all the visible content of the webpage.
<header>	class="college-header", id="home"	Defines the header section containing the university and department information.
–<h1>	—	Defines the main top-level heading on the webpage.
<h2>	—	Defines the department heading, **Computer Engineering**.
<nav>	class="sidebar"	Defines the navigation section containing links to different sections of the webpage.
<a>	href, onclick	Creates navigation links to different sections of the same webpage.
<p>	—	Defines a paragraph of text.
<main>	—	Defines the main content area of the

TAGS USED	ATTRIBUTES USED	DESCRIPTION
		webpage.
<section>	id	Defines separate content sections such as Vision, Mission, PSO, Gallery, Courses, and Recruiters.
	—	Creates an unordered list.
	—	Defines an individual item in a list.
<script>	—	Contains JavaScript used for smooth and centered scrolling.

Q -1 : Using HTML and CSS, create a bouncing loader animation.

Output :



Practical A - Bouncing Loader Animation

Html code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical A - Bouncing Loader Animation</title>
  <!-- external css link -->
  <link rel="stylesheet" href="a_bouncing_loader.css">
</head>
<body>

  <h2>Bouncing Loader Animation</h2>

  <!-- bouncing loader wrapper -->
  <div class="loader-container">
    <div class="dot"></div>
    <div class="dot"></div>
    <div class="dot"></div>
  </div>

</body>
</html>
```

Css code :

```
body {
  font-family: sans-serif;
  display: flex;
  flex-direction: column;
  justify-content: center;
```

```
  align-items: center;
  height: 100vh;
  margin: 0;
  background-color: #f9f9f9;
}

h2 {
  margin-bottom: 30px;
}

/* bouncing loader wrapper */
.loader-container {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 10px;
}

/* individual dots */
.dot {
  width: 16px;
  height: 16px;
  background-color: #3498db;
  border-radius: 50%;
  /* added bounce keyframes */
  animation: bounce 0.6s infinite alternate;
}

/* adding animation delays so dots bounce one after another */
.dot:nth-child(2) {
  animation-delay: 0.2s;
}

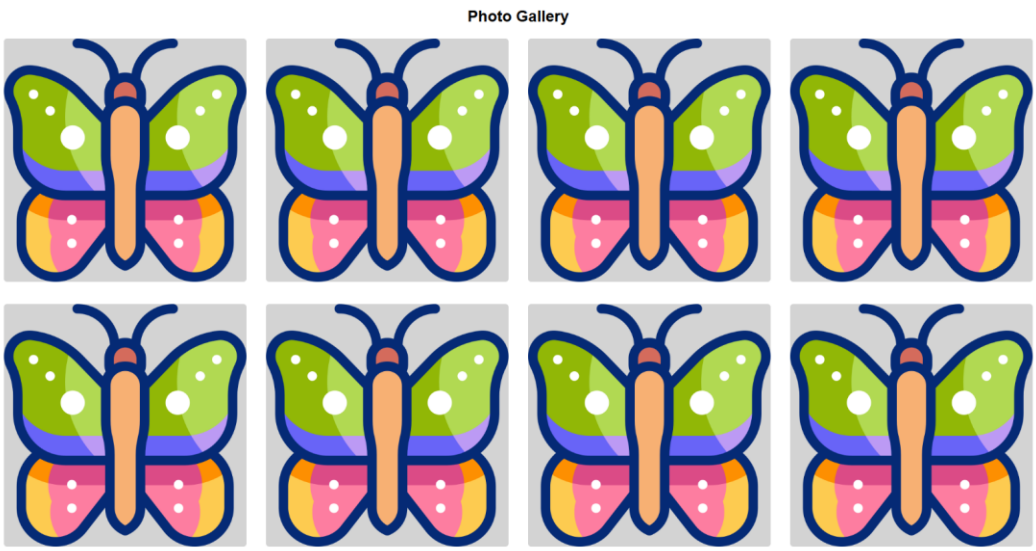
.dot:nth-child(3) {
  animation-delay: 0.4s;
```

```
}

/* keyframe animation moving dots up and down */
@keyframes bounce {
  from {
    transform: translateY(0);
    opacity: 1;
  }
  to {
    transform: translateY(-18px);
    opacity: 0.3;
  }
}
```

Q-2 : Create a photo gallery with CSS Flex.

Output:



Practical B - Photo Gallery using Flexbox

Html code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical B - Photo Gallery using Flexbox</title>
  <!-- external css link -->
  <link rel="stylesheet" href="b_photo_gallery.css">
</head>
<body>

  <h2>Photo Gallery</h2>

  <!-- gallery flex container using butterfly_3.png -->
  <div class="gallery">
    <div class="photo"><img src="butterfly_3.png" alt="Photo 1"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 2"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 3"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 4"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 5"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 6"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 7"></div>
    <div class="photo"><img src="butterfly_3.png" alt="Photo 8"></div>
  </div>

</body>
</html>
```

Css code:

```
body {
  font-family: sans-serif;
  padding: 20px;
}

h2 {
  text-align: center;
}

/* photo gallery container using flex */
.gallery {
  display: flex;
  gap: 30px;
  flex-wrap: wrap;
  justify-content: center;
}

.photo {
  width: 20%;
}

.photo img {
  width: 100%;
  background-color: lightgray;
  border-radius: 5px;
  transition: transform 0.2s ease;
}

.photo img:hover {
  transform: scale(1.05);
}

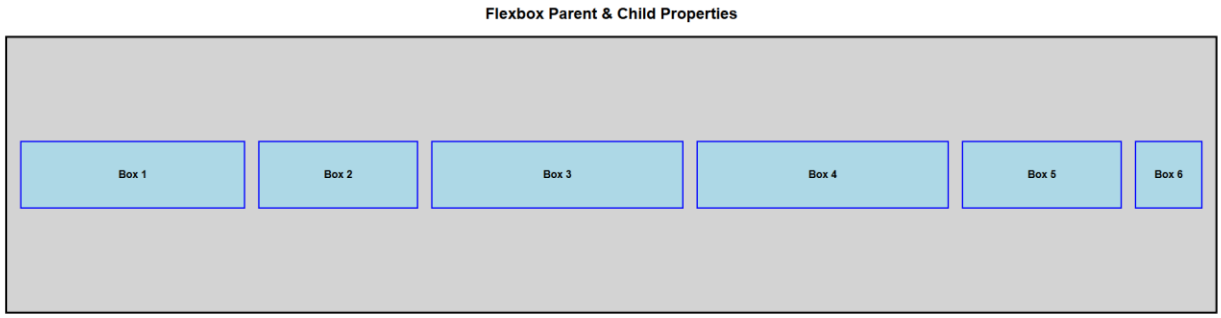
/* responsiveness for smaller screens */
```

```
@media (max-width: 700px) {
  .photo {
    width: 40%;
  }
}

@media (max-width: 400px) {
  .photo {
    width: 80%;
  }
}
```

Q-3 : Apply the Flex property on flex container and flex child

Output:



Html code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical C - Flex Container and Flex Child Properties</title>
  <!-- external css link -->
  <link rel="stylesheet" href="c_flex_container_child.css">
</head>
<body>

  <h2>Flexbox Parent & Child Properties</h2>

  <!-- flex parent container -->
  <div class="container">
    <!-- flex child items -->
    <div class="box box1">Box 1</div>
    <div class="box box2">Box 2</div>
    <div class="box box3">Box 3</div>
    <div class="box box4">Box 4</div>
    <div class="box box5">Box 5</div>
    <div class="box box6">Box 6</div>
  </div>

</body>
</html>
```

Css code :

```
body {
  font-family: sans-serif;
  padding: 20px;
```

```
}

h2 {
  text-align: center;
}

/* flex parent container properties */
.container {
  display: flex;
  flex-direction: row;
  justify-content: flex-start;
  align-items: center;
  flex-wrap: wrap;
  align-content: space-around;
  background-color: lightgrey;
  border: 3px solid black;
  height: 400px;
  padding: 10px;
}

/* base styling for flex children */
.box {
  width: 100px;
  height: 100px;
  margin: 10px;
  background-color: lightblue;
  border: 2px solid blue;
  text-align: center;
  line-height: 100px;
  font-weight: bold;
}

/* individual flex child properties */
.box1 {
  flex-grow: 1;
```

```
  flex-basis: 200px;
  flex-shrink: 10;
}

.box2 {
  flex-grow: 1;
  align-self: flex-start; /* overrides parent align-items */
}

.box3 {
  flex-grow: 2;
}

.box4 {
  flex-grow: 2;
  align-self: flex-end; /* overrides parent align-items */
}

.box5 {
  flex-grow: 1;
}

.box6 {
  flex-shrink: 9;
}
```